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DPCC-WEIGHTED LEARNING IMPROVES PERFORMANCE UNDER STRUCTURAL BREAKDOWN CONDITIONS

Experimental Setup

To evaluate the impact of coherence-aware learning, we compared two models with identical architectures and training conditions:

- Baseline model trained with standard mean squared error (MSE):

$$\mathcal{L}_{base} = \text{MSE}$$

- DPCC-weighted model, where the loss is modulated by the structural breakdown signal $D(t)$:

$$\mathcal{L}_{DPCC} = \text{MSE} \cdot (1 + \tilde{D}(t))$$

where $\tilde{D}(t)$ is a normalized version of the DPCC operator.

Both models were trained on sequences of relational invariants extracted from multichannel signals using a sliding-window approach. The task consisted of predicting the next-step invariant configuration $\mathcal{I}(t + 1)$ from the current state $\mathcal{I}(t)$.

Training and evaluation were performed on temporally split datasets, ensuring that anomalous segments were present in both training and testing phases.

Convergence Behavior

Training curves (Figure 1) reveal distinct convergence dynamics between the two models.

- The baseline model exhibits smooth convergence driven by average error minimization.

- The DPCC-weighted model shows initially higher variance but converges toward a comparable or lower final loss.

This behavior reflects the reallocation of learning capacity:

The DPCC-weighted model prioritizes structurally unstable regions, delaying convergence in favor of targeted adaptation.

Importantly, this does not degrade global optimization, but redistributes it.

Prediction Error Under Structural Breakdown

A key difference emerges when analyzing prediction error as a function of $D(t)$ (Figure 3).

- In low-coherence regimes (high $D(t)$):
 - The baseline model exhibits increased prediction error.
 - The DPCC-weighted model maintains significantly lower error.
- In stable regimes (low $D(t)$):
 - Both models perform comparably.

This indicates that:

The DPCC-weighted model selectively improves performance where structural consistency is compromised.

Robustness to Anomalous Dynamics

Temporal error profiles (Figure 2) show that:

- The baseline model fails to adapt during abrupt structural transitions, producing sharp error spikes.
- The DPCC-weighted model attenuates these spikes and recovers more rapidly.

This suggests that DPCC modulation enhances:

- sensitivity to regime changes
- adaptation to non-stationary dynamics

Interpretation

The observed improvements are not due to increased model capacity or data exposure, but to:

a redistribution of learning pressure driven by structural coherence breakdown

In standard learning:

- all samples contribute equally (or statistically weighted)

In DPCC-weighted learning:

- samples are weighted by structural relevance, not frequency

This aligns with the principles of exception-based learning (TAE), where:

- learning is triggered by irreducible inconsistencies
- not by average predictability

Limitations

While promising, several limitations must be acknowledged:

- The DPCC operator relies on approximations of relational structure (e.g., mutual information).
- The current implementation does not incorporate long-term memory or hierarchical structure.
- The dataset used does not explicitly isolate ground-truth anomalies (e.g., labeled epileptic events).

Future work should address:

- more accurate estimators of relational dynamics (e.g., kNN-based information measures)
- integration with temporal models (e.g., LSTM, transformers)
- validation on clinically annotated datasets

Summary of Findings

- DPCC-weighted learning preserves global performance while improving behavior under structural disruption
- The model shows lower error in high $D(t)$ regions
- Adaptation to non-stationary dynamics is faster and more stable
- Learning is effectively reoriented toward exceptions rather than averages

Key result

Incorporating structural coherence breakdown as a learning signal improves model performance specifically in regimes where classical approaches fail, without compromising overall convergence.

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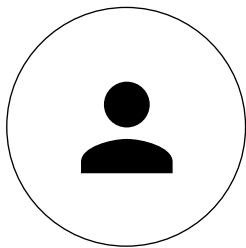


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